Construction of a Real-World Observational Dataset for Causal Inference Validation

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1 Introduction

The primary objective of this project is to evaluate causal inference techniques on both synthetic and real-world observational datasets. The synthetic dataset was intentionally designed with known treatment effects to benchmark different causal estimation methods under controlled conditions. While this provides a reliable environment for validating methodological correctness, practical applications of causal inference require analysis on real observational data where the true causal effects are unknown.

To bridge this gap, a real-world COVID-19 dataset was constructed using publicly available data sources. The purpose of this dataset is not to replace the synthetic benchmark but to validate whether the same causal inference pipeline can produce meaningful estimates under realistic observational settings.

Unlike the synthetic dataset, the real-world dataset contains naturally occurring treatment assignments, heterogeneous country-level characteristics, and temporal variation in government interventions. Consequently, the analysis becomes closer to practical decision-making scenarios commonly encountered in epidemiology, public policy, and healthcare analytics.

The overall workflow therefore consists of two complementary stages:

1. Validation of causal inference algorithms using a synthetic benchmark where the ground truth is known.

2. Application of the same causal inference framework to a carefully curated real-world observational dataset.

This document focuses on the methodology adopted for constructing the real-world dataset and explains the rationale behind every preprocessing decision before performing causal inference.

2 Motivation for Real-World Validation

Synthetic datasets play an important role in the development and evaluation of causal inference methods because they allow complete control over the underlying data-generating mechanism. Since the true treatment effects are known by construction, they provide an ideal environment for benchmarking different causal estimation techniques such as Propensity Score Matching (PSM), Inverse Probability Weighting (IPW), Augmented Inverse Probability Weighting (AIPW), and Double Machine Learning (DML). Any deviation between the estimated effect and the true effect can be directly quantified, making it possible to evaluate estimator bias, variance, and overall performance under controlled experimental conditions.

However, synthetic datasets cannot fully capture the complexity of real-world observational data. In practice, treatment assignments are rarely randomized and are influenced by multiple demographic, healthcare, socioeconomic, and policy-related factors. These naturally occurring confounders introduce systematic biases that make causal effect estimation significantly more challenging. Furthermore, real-world datasets contain measurement errors, reporting delays, heterogeneous populations, and missing observations, all of which must be addressed before meaningful causal conclusions can be drawn.

For this reason, synthetic benchmarking alone is insufficient to demonstrate the practical applicability of a causal inference framework. A robust evaluation requires validating the methodology on real observational data where the true treatment effects are unknown. Rather than measuring agreement with a known ground truth, the objective shifts towards obtaining statistically credible and scientifically interpretable causal estimates under realistic conditions.

In this project, the synthetic dataset serves as a methodological benchmark, while the real-world dataset acts as an external validation of the complete causal inference pipeline. The intention is not to compare the numerical treatment effects obtained from the two datasets, since they represent different data-generating processes. Instead, the objective is to evaluate whether the same causal methodology remains applicable when transitioning from a controlled simulation environment to naturally occurring observational data.

This two-stage validation strategy improves the reliability of the study. The synthetic benchmark demonstrates that the estimation procedures behave correctly under known conditions, whereas the real-world analysis demonstrates that the same methodology can be applied to practical decision-making problems where the underlying causal effects must be inferred directly from observational evidence.

3 Real-World Data Sources

To validate the causal inference framework developed using the synthetic benchmark dataset, publicly available national-level COVID-19 datasets were collected from two widely recognised sources. The selection of these datasets was guided by three primary objectives:

1. Availability of variables that closely resemble the synthetic benchmark.
2. Reliable and regularly updated country-level observations.
3. Suitability for observational causal inference analysis.

The final analytical dataset was constructed by integrating information from the following sources.

3.1 Our World in Data (OWID)

Our World in Data (OWID) is an internationally recognised open-access repository that compiles COVID-19 statistics from multiple official organisations, including the World Health Organization (WHO), national health agencies, the United Nations, and the World Bank. The dataset provides harmonised country-level indicators covering epidemiological outcomes, vaccination statistics, demographic characteristics, healthcare infrastructure, and socioeconomic indicators.

For this study, OWID serves as the primary source of both the outcome variable and several important confounding variables. The following information was extracted from the OWID dataset:

- Weekly COVID-19 new cases per million population
- Vaccination coverage
- Population density
- Hospital beds per thousand population
- Median age
- Percentage of population aged 65 years or older
- Gross Domestic Product (GDP) per capita

These variables collectively describe the demographic, healthcare, and socioeconomic characteristics that may simultaneously influence government intervention policies as well as disease transmission, thereby acting as potential confounders in the causal analysis.

3.2 Oxford COVID-19 Government Response Tracker (OxCGRT)

The Oxford COVID-19 Government Response Tracker (OxCGRT), developed by the Blavatnik School of Government at the University of Oxford, systematically records government policy responses implemented during the COVID-19 pandemic across countries.

The dataset contains multiple policy indicators summarised into the **Stringency Index**, which quantifies the intensity of non-pharmaceutical interventions such as:

- School closures
- Workplace closures
- Restrictions on public gatherings
- Public transport closures
- Stay-at-home requirements
- Domestic and international travel restrictions

The Stringency Index was selected as the treatment variable because it provides a continuous measure of government intervention intensity and directly represents the policy decisions whose causal impact on disease spread is of interest.

3.3 Consideration of Google Community Mobility Reports

Google Community Mobility Reports were initially considered as an additional data source because they provide behavioural information regarding changes in population mobility across different categories such as workplaces, retail locations, parks, and public transport.

However, after careful examination, the dataset was not incorporated into the final analytical panel. The primary reasons were:

1. Extensive sub-national reporting leading to multiple observations per country and date.
2. Considerable heterogeneity in geographical coverage across countries.
3. Increased preprocessing complexity due to regional duplication and inconsistent reporting.
4. The selected OWID and OxCGRT variables already captured the principal demographic, healthcare, socioeconomic, vaccination, and policy-related confounders required for the intended causal analysis.

Therefore, to maintain a clean, reproducible, and country-level observational panel, the final dataset was constructed exclusively using OWID and OxCGRT.

3.4 Rationale for Dataset Selection

The selected datasets complement one another in a manner consistent with the causal framework adopted in this project. While OWID provides reliable epidemiological outcomes together with demographic and healthcare covariates, OxCGRT captures government intervention policies that serve as the treatment of interest.

Together, these datasets provide sufficient information to estimate the causal effect of government intervention intensity on COVID-19 case incidence while appropriately adjusting for important observable confounding factors.

4 Construction of the Final Analytical Dataset

The raw datasets obtained from OWID and OxCGRT underwent a systematic preprocessing pipeline before being used for causal inference. The objective of this preprocessing stage was to construct a clean country-level observational panel that closely resembles the structure of the previously developed synthetic benchmark dataset while preserving the integrity of the original observations.

The preprocessing workflow consisted of the following stages.

4.1 Selection of Relevant Variables

Although both OWID and OxCGRT contain a large number of variables, only those directly relevant to the causal framework were retained. Variables were selected based on their theoretical relationship with the treatment assignment, disease transmission, healthcare capacity, and demographic characteristics.

From the OWID dataset, the following variables were extracted:

- Country
- Date
- New COVID-19 Cases per Million
- Percentage of Population Vaccinated
- Population Density
- Hospital Beds per Thousand
- Median Age
- Population Aged 65 Years or Older
- GDP per Capita

From the OxCGRT dataset, the following variables were retained:

- Country
- Date
- Government Stringency Index

Selecting only the required variables simplified the subsequent preprocessing steps while ensuring that the final dataset contained only variables with meaningful causal interpretation.

4.2 Temporal Filtering

The study was restricted to observations from the year 2021. This period was selected because it represents a phase of the pandemic during which vaccination campaigns had commenced globally while governments continued to implement varying levels of non-pharmaceutical interventions.

Restricting the analysis to a single calendar year provides a relatively homogeneous observational period and avoids combining fundamentally different pandemic phases such as the initial outbreak in 2020 with later vaccination periods.

4.3 Dataset Integration

The OWID and OxCGRT datasets were merged using two common identifiers:

- Country
- Date

An inner join was performed to ensure that only observations available in both datasets were retained. This guarantees that every observation simultaneously contains information regarding government interventions, epidemiological outcomes, demographic characteristics, and healthcare indicators.

Following the merge, duplicate country-date observations were checked and none were identified, confirming that every country-date pair appears exactly once in the integrated dataset.

4.4 Handling Missing Values

Different variables required different missing-value strategies depending on their nature.

Variables such as population density, median age, hospital beds, GDP per capita, and the proportion of elderly individuals represent relatively stable country-level characteristics that remain nearly constant over the study period. Missing values for these variables were first imputed using forward-fill and backward-fill operations within each country whenever partial information was available.

Countries that still contained missing structural variables after this step were considered incomplete for causal analysis.

Vaccination coverage exhibited missing observations primarily during the early stages of vaccine rollout. These missing values were interpreted as the absence of vaccination coverage and were therefore replaced with zero.

Observations with missing outcome values (new COVID-19 cases per million) were removed because they represented only a very small proportion of the overall dataset.

4.5 Country Quality Assessment

After handling missing values, a country-level quality assessment was performed.

For each country, the average proportion of missing values across all structural covariates was calculated. Countries with excessive missingness were excluded from further analysis to avoid introducing unnecessary uncertainty into the causal estimates.

A threshold of 10% average missingness was adopted. Countries exceeding this threshold were removed from the analytical dataset.

This quality-control procedure reduced the dataset from 180 countries to a final set of 52 countries with complete information across all selected variables.

The resulting dataset contained no remaining missing values.

4.6 Weekly Aggregation

The original datasets provide daily observations. However, daily COVID-19 reporting often contains substantial variability caused by reporting delays, weekend effects, and administrative inconsistencies.

To obtain more stable outcome measurements, the daily observations were aggregated into weekly country-level observations.

For each country-week combination:

- Weekly mean new cases per million was computed.
- Weekly mean vaccination coverage was computed.
- Weekly mean Stringency Index was computed.
- Static demographic and healthcare variables were retained unchanged.

This aggregation substantially reduced short-term reporting noise while preserving the overall temporal dynamics of the pandemic.

The first incomplete week spanning December 2020 was removed, resulting in a balanced panel consisting of exactly 52 complete epidemiological weeks for each country.

4.7 Construction of the Treatment Variable

The Stringency Index obtained from OxCGRT is originally measured on a continuous scale ranging from 0 to 100.

To maintain consistency with the synthetic benchmark dataset, where treatment was represented as a binary intervention, the continuous Stringency Index was transformed into a binary treatment variable.

Rather than selecting an arbitrary threshold, the median weekly Stringency Index across all observations was used as the cutoff value.

Formally,

$$Treatment = \begin{cases} 1, & \text{if Stringency Index} \geq \text{Median} \\ 0, & \text{otherwise} \end{cases}$$

Using the median threshold produced an almost perfectly balanced treatment assignment, with approximately 50% of observations belonging to the treated group and 50% belonging to the control group.

Balanced treatment groups improve overlap between treated and untreated observations, thereby facilitating more reliable estimation using propensity score methods and inverse probability weighting.

5 Variable Description and Correspondence with the Synthetic Benchmark

The primary objective of constructing the real-world dataset was not to exactly reproduce every variable present in the synthetic simulation, but rather to identify real-world observable variables that capture the same underlying causal mechanisms.

The synthetic benchmark dataset was generated using a mechanistic epidemiological simulation (SEIR framework), where several latent variables such as transmission rate and contact intensity were directly available. In contrast, real-world observational datasets do not provide direct measurements of these latent epidemiological quantities. Therefore, carefully selected observable proxies were used whenever direct measurements were unavailable.

Table 1 presents the correspondence between the synthetic benchmark variables and their real-world counterparts.

Table 1: Correspondence between Synthetic and Real-World Variables

Synthetic Variable	**Real-World Variable**	**Justification**
Treatment	Government Stringency Index	Government interventions directly influence disease transmission and represent the policy whose causal effect is being estimated.
Peak Infection	Weekly New Cases per Million	Represents the observed disease burden and serves as the outcome variable.
Vaccination Rate	People Vaccinated per Hundred	Measures immunity acquired through vaccination and influences infection dynamics.
Population Density	Population Density	Higher density increases opportunities for disease transmission.
Hospital Capacity	Hospital Beds per Thousand	Represents healthcare infrastructure available within a country.
Median Age	Median Age	Captures demographic differences that influence disease spread and severity.
Elderly Population	Population Aged 65+	Represents vulnerability of the population to COVID-19.
Socioeconomic Status	GDP per Capita	Captures differences in national economic development and healthcare resources.
Transmission Rate (β)	Government Stringency Index (Proxy)	The transmission rate cannot be directly observed in real-world datasets. Government restrictions substantially influence contact patterns and therefore act as a practical proxy for transmission intensity.
Contact Intensity	Government Stringency Index (Proxy)	Strict government interventions reduce interpersonal interactions, making Stringency Index an indirect measure of average contact intensity.
Mask Compliance	Government Stringency Index (Partial Proxy)	Although mask usage is not directly available in the selected datasets, stricter govern-

5.1 Description of Individual Variables

Each variable included in the final analytical dataset serves a specific role within the causal framework.

- **Country**
  Identifies each observational unit in the panel dataset. Every country contributes one observation for each epidemiological week.

- **Week**
  Represents the temporal dimension of the panel dataset after aggregating daily observations into weekly averages.

- **Treatment**
  A binary indicator derived from the weekly Government Stringency Index. Countries with weekly Stringency Index greater than or equal to the median value are considered treated, while the remaining observations form the control group.

- **Stringency Index**
  Measures the intensity of government interventions implemented to reduce COVID-19 transmission. This variable forms the basis for defining the treatment assignment.

- **Weekly New Cases per Million**
  Represents the average number of newly reported COVID-19 cases per million population during each epidemiological week. This is the primary outcome variable whose causal relationship with government interventions is investigated.

- **Vaccination Rate**
  Represents the percentage of individuals vaccinated within a country. Vaccination directly affects disease transmission and therefore acts as an important confounder.

- **Population Density**
  Represents the number of individuals living per square kilometre. Higher density increases opportunities for disease spread and therefore influences both treatment decisions and disease burden.

- **Hospital Beds per Thousand**
  Represents healthcare infrastructure available within a country. Countries with stronger healthcare systems often adopt different intervention strategies and may exhibit different disease outcomes.

- **Median Age**
  Represents the age structure of the country's population. Population age influences both disease susceptibility and government policy decisions.

- **Population Aged 65 Years or Older**
  Represents the proportion of elderly individuals within the population. Older populations generally face higher disease risk and may motivate stronger policy interventions.

- **GDP per Capita**
  Acts as a socioeconomic indicator reflecting national wealth, healthcare investment, and resource availability. It is treated as a confounding variable because it may simultaneously influence intervention policies and disease outcomes.

5.2 Final Analytical Dataset

Following preprocessing, filtering, weekly aggregation, and treatment construction, the final observational panel consists of:

- 52 countries
- 52 complete epidemiological weeks
- 2704 country-week observations
- No duplicate observations
- No missing values
- Balanced binary treatment groups (approximately 50% treated and 50% control)

This final dataset closely mirrors the structure of the synthetic benchmark while retaining the complexity and heterogeneity naturally present in real-world observational data. Consequently, it provides an appropriate platform for validating the complete causal inference pipeline under realistic conditions.

6 Planned Causal Inference Framework

After constructing the final observational panel dataset, the next stage of the project focuses on estimating the causal effect of government intervention policies on COVID-19 disease burden. Unlike traditional predictive modelling, the objective is not merely to forecast infection counts but to estimate how infection outcomes would have changed under different policy interventions while appropriately adjusting for confounding factors.

6.1 Research Question

The central research question of this study is:

> *"What is the causal effect of stringent government interventions on the weekly number of COVID-19 cases per million population after adjusting for demographic, healthcare, vaccination, and socioeconomic confounding factors?"*

This question aims to move beyond correlation and estimate the effect that government interventions have on disease transmission under observational conditions.

6.2 Definition of Treatment, Outcome and Confounders

The causal analysis is formulated using the following components.

Treatment

The treatment variable is a binary indicator representing the intensity of government intervention.

$$Treatment = \begin{cases} 1, & \text{High Government Stringency} \\ 0, & \text{Low Government Stringency} \end{cases}$$

The treatment assignment is obtained by thresholding the weekly Government Stringency Index using its median value, producing nearly balanced treated and untreated groups.

Outcome

The outcome variable is

$$\text{Weekly New COVID-19 Cases per Million Population}$$

which measures the disease burden observed during each epidemiological week.

Observed Confounders

The following variables are treated as observed confounders because they may simultaneously influence both government intervention policies and COVID-19 transmission.

- Vaccination Rate
- Population Density
- Hospital Beds per Thousand
- Median Age
- Population Aged 65 Years or Older
- GDP per Capita

Adjusting for these variables is intended to reduce confounding bias when estimating causal effects.

6.3 Conceptual Causal Model

The assumed causal structure is based on epidemiological and public health knowledge.

Government intervention policies directly influence disease transmission by restricting mobility, limiting social interactions, and reducing opportunities for viral spread. However, policy decisions are themselves influenced by demographic characteristics, healthcare infrastructure, vaccination coverage, and socioeconomic conditions. These variables therefore act as confounders because they affect both treatment assignment and the observed outcome.

A Directed Acyclic Graph (DAG) will be constructed to explicitly represent these causal assumptions before causal effect estimation. The DAG will also serve as the basis for identifying valid adjustment sets within the DoWhy framework.

6.4 Planned Estimation Strategy

To ensure robust causal estimation, multiple complementary causal inference methods will be applied.

1. Propensity Score Matching (PSM)

Propensity Score Matching will be used to construct treated and untreated groups with similar observed characteristics.

The primary objective is to estimate the Average Treatment Effect on the Treated (ATT) while reducing selection bias introduced by non-random treatment assignment.

Balance diagnostics will be evaluated before and after matching to assess the quality of covariate balance.

2. Inverse Probability Weighting (IPW)

Inverse Probability Weighting creates a weighted pseudo-population in which treatment assignment becomes independent of observed confounders.

This approach enables estimation of the Average Treatment Effect (ATE) using all available observations rather than only matched pairs.

3. Augmented Inverse Probability Weighting (AIPW)

AIPW combines propensity score weighting with outcome regression, producing a doubly robust estimator.

This estimator remains consistent provided that either the propensity score model or the outcome regression model is correctly specified, thereby improving robustness against model misspecification.

4. DoWhy Framework

The DoWhy framework will be used to formalise the complete causal analysis.

The analysis consists of four stages:

1. Specification of the causal graph.
2. Identification of the causal estimand.
3. Estimation of the causal effect.
4. Refutation and robustness testing.

Refutation tests such as placebo treatments, random common causes, and subset validation will be performed to evaluate the stability of the estimated causal effects.

5. EconML

Finally, heterogeneous treatment effects will be estimated using the EconML library.

Methods such as Double Machine Learning (DML) and Causal Forests will be employed to estimate Conditional Average Treatment Effects (CATEs), allowing investigation of whether government interventions have different effects across countries with different demographic and healthcare characteristics.

6.5 Relationship Between Synthetic and Real-World Analyses

The synthetic benchmark and the real-world analysis serve complementary purposes.

The synthetic dataset provides an environment where the true treatment effects are known, enabling quantitative evaluation of estimator bias and accuracy.

In contrast, the real-world dataset represents an observational setting where the true causal effects are unknown. Consequently, the objective is not to reproduce the synthetic results but to examine whether the same causal inference methodology produces statistically credible and scientifically interpretable estimates under realistic conditions.

Together, these two stages provide a comprehensive evaluation of the proposed causal inference framework by demonstrating both methodological validity under controlled simulation and practical applicability in real-world observational data.

7 Overall Workflow of the Study

The overall methodology adopted in this project consists of two complementary stages. The first stage focuses on validating causal inference methods using a synthetic benchmark dataset where the true treatment effects are known. The second stage applies the same causal inference framework to real-world observational data in order to evaluate its practical applicability.

Figure 1 illustrates the complete workflow followed throughout this project.

Raw Public Datasets (OWID + OxCGRT)

$\downarrow$

Variable Selection

$\downarrow$

Merge by Country and Date

$\downarrow$

Missing Value Treatment

$\downarrow$

Country Quality Assessment

$\downarrow$

Weekly Aggregation

$\downarrow$

Construction of Binary Treatment

$\downarrow$

Final Observational Panel

$\downarrow$

Exploratory Causal Analysis

$\downarrow$

Propensity Score Estimation

$\downarrow$

Propensity Score Matching (PSM)

$\downarrow$

Inverse Probability Weighting (IPW)

$\downarrow$

Augmented IPW (AIPW)

$\downarrow$

DoWhy Framework

$\downarrow$

EconML (DML + Causal Forest)

$\downarrow$

Comparison with Synthetic Benchmark

Figure 1: Overall workflow adopted for the real-world causal inference study.

The workflow follows the standard causal inference pipeline beginning with careful dataset construction, followed by causal effect estimation using multiple complementary methodologies. The synthetic benchmark provides methodological validation, whereas the real-world analysis evaluates the practical robustness of the proposed causal framework.

8 Conceptual Directed Acyclic Graph (DAG)

A Directed Acyclic Graph (DAG) is used to formally represent the assumed causal relationships between the treatment, outcome, and observed confounding variables. Rather than constructing the DAG arbitrarily, its structure is derived from established epidemiological knowledge regarding COVID-19 transmission and public health policy.

The proposed DAG assumes that government intervention policies directly influence disease transmission by reducing opportunities for interpersonal contact. At the same time, government decisions regarding intervention intensity are themselves influenced by demographic characteristics, healthcare infrastructure, vaccination coverage, and socioeconomic conditions.

Consequently, these variables simultaneously affect both treatment assignment and disease burden and therefore satisfy the definition of confounders.

The principal causal assumptions are summarised below.

- Population density increases opportunities for interpersonal interaction and therefore influences disease transmission.
- Vaccination coverage reduces susceptibility to infection and consequently decreases disease burden.
- Countries with stronger healthcare infrastructure may adopt different intervention strategies and also experience different disease outcomes.
- Demographic variables such as median age and the proportion of elderly individuals influence disease severity and government decision-making.
- GDP per capita reflects national healthcare resources and administrative capacity, affecting both intervention policies and observed infection rates.
- Government Stringency Index directly influences disease transmission through restrictions on mobility and public interaction.

Based on these assumptions, the causal question investigated in this study is:

Does stronger government intervention reduce COVID-19 disease burden after adjusting for observed confounding variables?

The corresponding DAG will be implemented within the DoWhy framework to identify valid adjustment sets before estimating causal effects.

dag.png

Figure 2: Directed Acyclic Graph representing the assumed causal relationships between treatment, outcome, and observed confounders.

9 Limitations and Future Scope

Although considerable effort was devoted to constructing a clean and reliable observational dataset, several limitations remain inherent to real-world causal inference studies.

Unlike the synthetic benchmark, where the complete data-generating mechanism is known, real-world observational datasets contain variables that cannot be directly measured. For example, the true disease transmission rate (β), individual contact intensity, behavioural compliance, and actual mask usage are latent epidemiological variables that are not publicly available at the country level. Consequently, observable policy indicators such as the Government Stringency Index were used as practical proxies to capture changes in transmission-related behaviour.

Furthermore, although important demographic, healthcare, and socioeconomic confounders were included in the analysis, unobserved confounding may still exist. Factors such as cultural behaviour, testing strategies, reporting accuracy, viral variants, healthcare accessibility, and public adherence to government policies cannot be completely controlled using publicly available datasets. Therefore, the estimated causal effects should be interpreted as conditional on the observed confounders included in the analysis.

Another limitation concerns the observational nature of the data. Since government interventions were not randomly assigned, treatment assignment may still depend on unknown factors. Methods such as Propensity Score Matching, Inverse Probability Weighting, Augmented Inverse Probability Weighting, and Double Machine Learning attempt to reduce this bias, but they cannot completely eliminate bias arising from unmeasured confounding.

Future work may incorporate additional behavioural and mobility datasets, more detailed regional information, alternative policy indicators, and sensitivity analyses for hidden confounding. Incorporating epidemiological compartment models together with observational causal inference may also provide a stronger integration between simulation-based and real-world analyses.

Despite these limitations, the proposed framework demonstrates a systematic methodology for transitioning from a controlled synthetic benchmark to a real-world observational dataset while maintaining a consistent causal inference pipeline.

A Appendix A: Final Real-World Dataset Summary

The preprocessing pipeline resulted in a clean observational panel dataset suitable for causal inference analysis.

Table 2: Summary of the Final Analytical Dataset

Characteristic	**Value**
Countries	52
Complete Epidemiological Weeks	52
Country-Week Observations	2704
Variables	11
Duplicate Observations	0
Missing Values	0
Treatment Balance	50.15% Treated
	49.85% Control
Outcome Variable	Weekly New Cases per Million
Treatment Variable	Binary Government Stringency

The final dataset represents a balanced country-level observational panel with complete information for every country-week combination included in the study.

B Appendix B: Variable Dictionary

Variable	**Description**
Country	Country corresponding to each observational unit.
Week	Weekly observation period after aggregating daily records.
Treatment	Binary treatment indicating whether weekly Government Stringency Index is greater than or equal to the median value.
Stringency Index	Government intervention intensity reported by OxCGRT.
Cases per Million	Weekly average COVID-19 new cases per million population.
Vaccination Rate	Weekly average percentage of vaccinated individuals.
Population Density	Population per square kilometre.

Hospital Beds	Hospital beds available per thousand population.
Median Age	Median age of the country's population.
Age 65+	Percentage of population aged 65 years or older.
GDP per Capita	Gross Domestic Product per capita.

C Appendix C: Summary of Data Preprocessing

The preprocessing workflow adopted in this study is summarised below.

1. Downloaded publicly available COVID-19 datasets from OWID and OxCGRT.
2. Selected only variables relevant to the causal framework.
3. Filtered observations to the year 2021.
4. Merged datasets using Country and Date as common identifiers.
5. Removed duplicate observations.
6. Treated missing values according to variable characteristics.
7. Performed country-level quality assessment and excluded countries with excessive missing structural information.
8. Aggregated daily observations into weekly country-level observations.
9. Removed the incomplete first epidemiological week.
10. Constructed a binary treatment variable using the median weekly Government Stringency Index.
11. Obtained the final balanced observational panel consisting of 52 countries and 2704 observations.